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B.Tech. Degree III Semester Regular/Supplementary Examination January 2023

CE/CS/EC/EE/IT/ME/SE 19-200-0301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES
(2019 Scheme)

Maximum Marks: 60

Time: 3 Hours

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve linear system of equations and to determine eigen values and eigen vectors of a matrix.
CO2: Understand the concept of vector space and subspace.

CO3: Determine Fourier series expansion of functions and transform.

CO4: Solve linear differential equation and integral equation using Laplace transform.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Find eigen values and eigen vectors of $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.	3	L2, L3	1	1
(b) Form quadratic form corresponding to $A = \begin{bmatrix} 5 & -3 \\ -3 & 8 \end{bmatrix}$.	3	L1, L2, L3	1	2
(c) Find a non zero vector W that is orthogonal to $(1, 3, 2)$ and $(1, 2, 3)$ in R^3 .	3	L3	2	2
(d) Find the value of λ for which the following set of vectors $(\lambda, 0, 2)$ $(2, 1, 0)$ $(0, 3, 6)$ form a basis of R^3 .	3	L3, L4	2	1
(e) Obtain the Fourier series expansion of $f(x)$ if $f(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ x & \text{for } 1 < x < 2 \end{cases}$ and $f(x+2) = f(x)$.	3	L2, L3	3	2
(f) Obtain half range sine series for e^x in $0 < x < 1$.	3	L2, L3	3	2
(g) Find the Laplace transform of $f(t)$ where $f(t) = \begin{cases} \cos t & 0 < t < \pi \\ \sin t & t > \pi \end{cases}$.	3	L3	4	2
(h) Find $L^{-1}\left(\frac{e^{-2s}}{s-3}\right)$.	3	L4	4	3

PART B

(4 × 12 = 48)

II. (a) Find the values of λ and μ for which the equation $x + y + z = 3$; $x + 2y + 2z = 6$; $x + \lambda y + 3z = \mu$ have (i) no solution (ii) unique solution (iii) more than one solution.	6	L4	1	2
(b) State Cayley-Hamilton theorem. Using this theorem find inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.	6	L2, L3	1	2

OR

(P.T.O.)

Marks	BL	CC
6	L2, L3	1

III. (a) Find the rank of the matrix.

$$A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 4 & 7 & 10 \\ 5 & 7 & 11 & 17 \\ 6 & 8 & 13 & 16 \end{bmatrix}$$

(b) Diagonalise the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix}$ by means of an orthogonal transformation. 6 L4 1 3

IV. (a) Let V consists of all real polynomials of degree ≤ 4 with usual polynomial addition and scalar multiplication. Let $W = \{\text{polynomials of degree } \leq 4 \text{ with coefficient of } x^2 = 0\}$. Then verify W is a subspace or not. 6 L4 2 2

(b) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a transformation defined by $T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ z \end{bmatrix}$. Verify T is a linear transformation or not. 6 L3 2 2

OR

V. (a) Find the dimension and basis of the vector space $V = \text{set of all } 2 \times 2 \text{ matrices}$. 6 L4 2 2

(b) Using Gram Schmidt orthonormalisation process find an orthogonal and orthonormal basis $V = P_2(x)$, $x \in [0,1]$ $B = \{1, x, x^2\}$ 6 L3 2 2

VI. (a) Find the Fourier series to represent the function $f(x) = \begin{cases} x & \text{for } 0 \leq x \leq \pi \\ 2\pi - x & \text{for } \pi \leq x \leq 2\pi \end{cases}$ and deduce that 6 L4, L3 3 2

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

(b) Find the Fourier cosine transform of e^{-x^2} . 6 L4 3 2

OR

VII. (a) Develop $x \sin x$ in half range cosine series in the range $0 < x < \pi$ 6 L4, L3 3 2

(b) Using Fourier integral show that 6 L4 3 2

$$\int_0^\infty \frac{1 - \cos \pi \lambda}{\lambda} \sin x \lambda d\lambda = \begin{cases} \frac{\pi}{2} & \text{in } 0 < x < \pi \\ 0 & \text{in } x > \pi \end{cases}$$

(Continued)

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VIII. (a) Solve the integral equation $y(t) = 1 + \int_0^t y(u) \sin(t-u) du$.

6 L4, 4 3
L5

(b) State convolution theorem. Find $L^{-1} \frac{1}{(s^2 + 4)^2}$ using convolution theorem.

6 L1, 4 2
L3

OR

IX. (a) (i) Prove the symmetric property of beta function.

3 L1, 4 2
L3

(ii) Define gamma function and prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

3 L1, 4 2
L3

(b) Solve the equation $y'' - 3y' + 2y = 4t + e^{3t}$ when $y(0) = 1$ and $y'(0) = -1$.

6 L4, 4 3
L5

Bloom's Taxonomy Levels

L1 = 6%, L2 = 12%, L3 = 38%, L4 = 40%, L5 = 4%.
